HDOC DAY-56

Write a C program that reads the size and elements of a variable-length integer array, arr[], and an integer k such that 0 < k ≤ size. Pass the array to a function named first_negative() using its address. Use pointers to pass the array between the calling function and first_negative(). The function first_negative() must:

- Consider every contiguous subarray of size k as a sliding window.
- Find the first negative integer in each window while moving from left to right.
- Display the first negative integer in the order of the corresponding windows.
- Display 0 if a window does not contain any negative integer.
- Separate the results by spaces.
- Display -1 if k is invalid.
- Use a nested-loop approach to solve the problem.

Example 1:

Array: 12 -1 -7 8 -15 30 16 28
k: 3

Output: -1 -1 -7 -15 -15 0

Example 2:

Array: 1 2 -3 4 5 -6
k: 2

Output: 0 -3 -3 0 -6

Example 3:

Array: 1 2 3 4
k: 2

Output: 0 0 0

Example 4:

Array: 1 2 3
k: 4

Output: -1

Algorithm:

1. Read size.
2. Declare the variable-length integer array arr[size].
3. Read all array elements.
4. Read k.
5. Pass the address of the array to first_negative().
6. If k <= 0 or k > size, print -1.
7. Otherwise, use an outer loop to generate every window of size k.
8. For each window, use an inner loop to find its first negative integer.
9. Print that negative integer and move to the next window.
10. If no negative integer exists in a window, print 0.

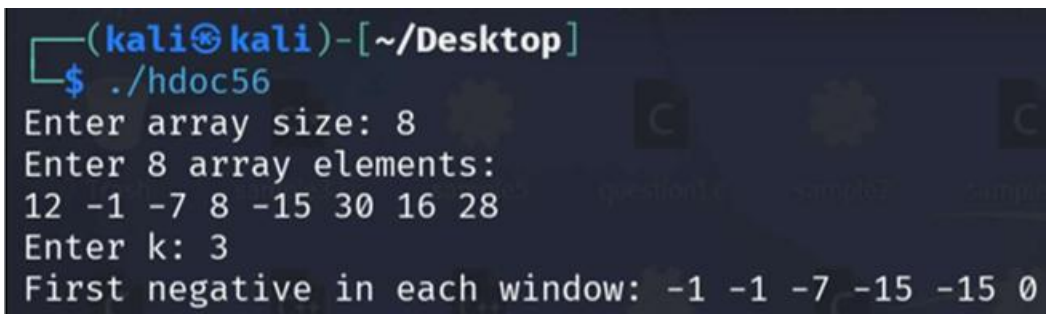
Test Case Table:

Test Case	Input array	k	Expected Output
1	12 -1 -7 8 -15 30 16 28	3	-1 -1 -7 -15 -15 0
2	1 2 3 4 5	2	0 0 0 0
3	-4 -2 -8 -1 -6	3	-4 -2 -8
4	3 -2 5 -1	1	0 -2 0 -1
5	3 2 5 -1	4	-2

Compilation and Execution:

The following program has been compiled and executed successfully.

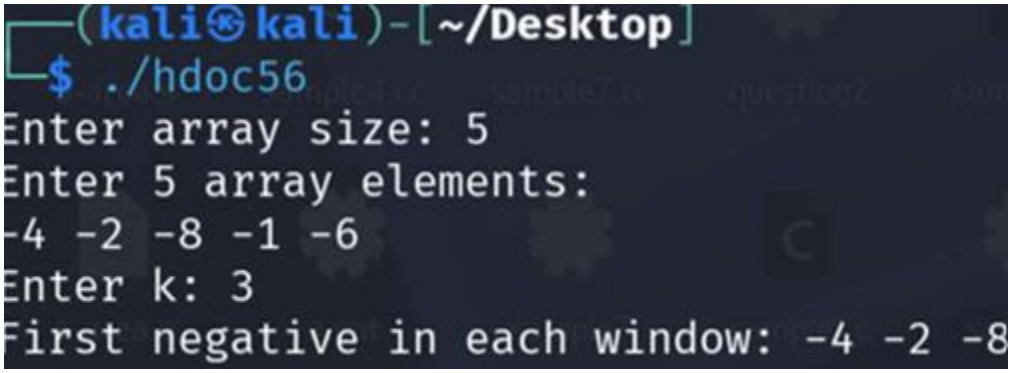
Test Case 1:



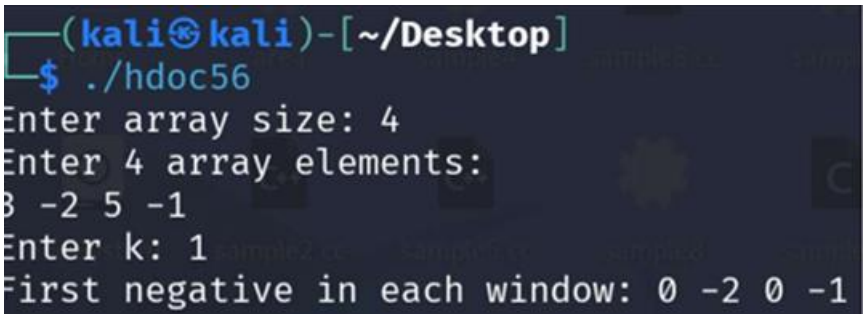
Test Case 2:



Test Case 3:



Test Case 4:



Test Case 5:

